SIEMENS

TIA Portal Openness: API for automation of engineering workflows

Manual

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1.1 Screens

1.1.1 Creating user-defined screen folders

Requirement

- The TIA Portal Openness application is connected to the TIA Portal.
  See Connecting to the TIA Portal
- A project is open.
  See Opening a project

Program code

Modify the following program code to create a user-defined screen folder:

```
Copies the following program code to clipboard.
//Creates a screen folder
private static void CreateScreenFolder(HmiTarget hmitarget)
{
    ScreenUserFolder myCreatedFolder =
hmitarget.ScreenFolder.Folders.Create("myScreenFolder");
}
```

1.1.2 Deleting a screen from a folder

Requirement

- The TIA Portal Openness application is connected to the TIA Portal.
  See Connecting to the TIA Portal
- A project is open.
  See Opening a project

Application

Note

You cannot delete a permanent area. A permanent area is a system screen that is always present.

Program code

Modify the following program code to delete a screen from a specific folder:

```
Copies the following program code to clipboard.

public static void DeleteScreenFromFolder(HmiTarget
hmiTarget)
{
    ScreenUserFolder screenUserFolder =
hmiTarget.ScreenFolder.Folders.Find("myScreenFolder");
    ScreenComposition screens = screenUserFolder.Screens;
    Screen screen = screens.Find("myScreenName");
    if (screen != null)
    {
        screen.Delete();
    }
}
```

1.1.3 Deleting a screen template from a folder

Requirement

- The TIA Portal Openness application is connected to the TIA Portal.
  See Connecting to the TIA Portal
- A project is open.
  See Opening a project
- The project contains an HMI device.

Program code

Modify the following program code to delete a screen template from a specific folder:

```
Copies the following program code to clipboard.

private static void DeleteScreenTemplateFromFolder(HmiTarget
hmiTarget)
```

```
{
    string templateName = "MyScreenTemplate";
    ScreenTemplateUserFolder folder =
hmiTarget.ScreenTemplateFolder.Folders.Find("myScreenTemplateFolder");
    ScreenTemplateComposition templates = folder.ScreenTemplates;
    ScreenTemplate template = templates.Find(templateName);
    if (template != null)
    {
        template.Delete();
    }
}
```

1.1.4 Deleting all screens from a folder

Requirement

- The TIA Portal Openness application is connected to the TIA Portal.
  See Connecting to the TIA Portal
- A project is open.
  See Opening a project

Application

Note

You cannot delete a permanent area. A permanent area is a system screen that is always present.

Program code

Modify the following program code to delete all screens from a specific folder:

```
 Copies the following program code to clipboard.
private static void DeleteAllScreensFromFolder(HmiTarget
hmitarget)
//Deletes all screens from a user folder or a system folder
{
```

```
    ScreenUserFolder folder =
hmitarget.ScreenFolder.Folders.Find("myScreenFolder");
    //or ScreenSystemFolder folder = hmitarget.ScreenFolder;
    ScreenComposition screens = folder.Screens;
    List<Screen> list = new List<Screen>();
    foreach(Screen screen in screens)
    {
        list.Add(screen);
    }
    foreach (Screen screen in list)
    {
        screen.Delete();
    }
}
```

1.2 Cycles

1.2.1 Deleting a cycle

Requirement

- The TIA Portal Openness application is connected to the TIA Portal.
  See Connecting to the TIA Portal
- A project is open.
  See Opening a project
- The project contains an HMI device.

Application

You cannot delete standard cycles.

You can identify whether cycles have actually been deleted based on the composition in the object model (composition count) of the respective cycle. It is no longer possible to access these cycles.

Program code

Modify the following program code to delete a cycle from an HMI device:



```
public static void DeleteCycle(HmiTarget hmiTarget)
{
    CycleComposition cycles = hmiTarget.Cycles;
    Cycle cycle = cycles.Find("myCycle");
    cycle.Delete();
}
```

1.3 Text lists

1.3.1 Deleting a text list

Requirement

- The TIA Portal Openness application is connected to the TIA Portal.
  See Connecting to the TIA Portal
- A project is open.
  See Opening a project
- The project contains an HMI device.

Program code

Modify the following program code to delete a selected text list and all associated list entries from an HMI device:




```
public static void DeleteTextList(HmiTarget hmiTarget)
{
    TextListComposition textLists = hmiTarget.TextLists;
    TextList textList = textLists.Find("myTextList");
    textList.Delete();
}
```

1.4 Graphic lists

1.4.1 Deleting a graphic list

Requirement

- The TIA Portal Openness application is connected to the TIA Portal.
  See Connecting to the TIA Portal

- A project is open.
  See Opening a project
- The project contains an HMI device.

Program code

Modify the following program code to delete a selected graphic list and all associated list entries from an HMI device:

```
Copies the following program code to clipboard.

private static void DeleteGraphicList(HmiTarget hmiTarget)
{
    GraphicListComposition graphicLists =
hmiTarget.GraphicLists;
    GraphicList graphicList =
graphicLists.Find("myGraphicList");
    graphicList.Delete();
}
```

1.5 Connections

1.5.1 Deleting a connection

Requirement

- The TIA Portal Openness application is connected to the TIA Portal.
  See Connecting to the TIA Portal
- A project is open.
  See Opening a project
- The project contains an HMI device.

Program code

Modify the following program code to delete a selected communication connection from an HMI device:

```
Copies the following program code to clipboard.

private static void DeleteConnection(HmiTarget hmiTarget)
{
```

```
    ConnectionComposition connections =
hmiTarget.Connections;
    Connection connection =
connections.Find("HMI_connection_1");
    connection.Delete();
}
```

1.6 Tag table

1.6.1 Creating user-defined folders for HMI tags

Requirement

- The TIA Portal Openness application is connected to the TIA Portal.
  See Connecting to the TIA Portal
- A project is open.
  See Opening a project

Program code

Modify the following program code to create a user-defined folder for HMI tags:

```
 Copies the following program code to clipboard.
private static void CreateUserfolderForHMITags(HmiTarget
hmitarget)
// Creates an HMI tag user folder
{
    TagSystemFolder folder = hmitarget.TagFolder;
    TagUserFolder myCreatedFolder =
folder.Folders.Create("MySubFolder");
}
```

1.6.2 Enumerating tags of an HMI tag table

Requirement

- The TIA Portal Openness application is connected to the TIA Portal.
  See Connecting to the TIA Portal
- A project is open.
  See Opening a project

Program code

Modify the following program code to enumerate all tags of an HMI tag table:

```
Copies the following program code to clipboard.
private static void EnumerateTagsInTagtable(HmiTarget
hmitarget)
// //Enumerates all tags of a tag table
{
    TagTable table =
hmitarget.TagFolder.TagTables.Find("MyTagtable");
    // Alternatively, you can access the default tag table:
    // TagTable defaulttable =
hmitarget.TagFolder.DefaultTagTable;

    TagComposition tagComposition = table.Tags;
    foreach (Tag tag in tagComposition)
    {
        // Add your code here
    }
}
```

1.6.3 Deleting an individual tag from an HMI tag table

Requirement

- The TIA Portal Openness application is connected to the TIA Portal.
  See Connecting to the TIA Portal
- A project is open.
  See Opening a project

Program code

Modify the following program code to delete a specific tag from an HMI tag table:

```
Copies the following program code to clipboard.
private static void DeleteATag(HmiTarget hmiTarget)
{
```

```
string tagName = "MyTag";
TagTable defaultTagTable =
hmiTarget.TagFolder.DefaultTagTable;
TagComposition tags = defaultTagTable.Tags;
Tag tag = tags.Find(tagName);
tag.Delete();
}
```

1.6.4 Deleting a tag table from a folder

Requirement

- The TIA Portal Openness application is connected to the TIA Portal.
  See Connecting to the TIA Portal
- A project is open.
  See Opening a project
- The project contains an HMI device.

Application

You cannot delete the default tag table

Program code

Modify the following program code:

```
Copies the following program code to clipboard.
// Delete a tag table from a specific folder
private static void DeleteTagTable(HmiTarget hmiTarget)
{
    string tableName = "myTagTable";
    TagSystemFolder tagSystemFolder = hmiTarget.TagFolder;
    TagTableComposition tagTables =
tagSystemFolder.TagTables;
    TagTable tagTable = tagTables.Find(tableName);
    tagTable.Delete();
}
```

1.7 VB scripts

1.7.1 Creating user-defined script folders

Requirement

- The TIA Portal Openness application is connected to the TIA Portal.
  See Connecting to the TIA Portal
- A project is open.
  See Opening a project

Program code

Modify the following program code to create a user-defined script subfolder below a system folder or another user-defined folder:

```
Copies the following program code to clipboard.
private static void CreateFolderInScriptfolder(HmiTarget hmitarget)
//Creates a script user subfolderVBScriptSystemFolder
{
    VBScriptSystemFolder vbScriptFolder = hmitarget.VBScriptFolder;
    VBScriptUserFolderComposition vbScriptFolders = vbScriptFolder.Folders;
    VBScriptUserFolder vbScriptSubFolder = vbScriptFolder.Folders.Create("mySubfolder");
}
```

1.7.2 Deleting a VB script from a folder

Requirement

- The TIA Portal Openness application is connected to the TIA Portal.
  See Connecting to the TIA Portal
- A project is open.
  See Opening a project
- The project contains an HMI device.

Program code

Modify the following program code to delete a VB script from a specific folder:

```
Copies the following program code to clipboard.
//Deletes a vbscript from a script folderVBScriptSystemFolder
private static void DeleteVBScriptFromScriptFolder(HmiTarget
hmitarget)
{
    VBScriptUserFolder vbscriptfolder =
hmitarget.VBScriptFolder.Folders.Find("MyScriptFolder");
    var vbScripts = vbscriptfolder.VBScripts;
    if (null != vbScripts)
    {
        var vbScript = vbScripts.Find("MyScript");
        vbScript.Delete();
    }
}
```

1.7.3 Deleting a user-defined folder of an HMI device

Requirement

- The TIA Portal Openness application is connected to the TIA Portal.

  See Connecting to the TIA Portal
- A project is open.

  See Opening a project

Program code

Modify the following program code to delete an user-defined folder of an HMI device:

```
Copies the following program code to clipboard.
HmiTarget hmiTarget = ...;
ScreenUserFolder screenUserGroup =
hmiTarget.ScreenFolder.Folders.Find("MyUserFolder");
screenUserGroup.Delete();
```